Search methods: benchmark results and cost

Every search method the simulated environment was tested with, and every optimization tried on its original Europe PMC + LitSense retrieval. Three studies, with different questions and metrics, so each has its own table.

1. The search methods on the repeated benchmark

413 questions per run (set A: 141 with a hidden true value; set B: 272 never stated), up to 10 runs, scored by a three-judge panel (Claude Haiku 5.5, GLM 5.3 Flash, GPT-6 Luna). Mean ± SD over runs. Every method uses medsim's stages on DeepSeek V4 Flash; only the retrieval differs. The shaded row is medsim's current default.

Search method	Runs	Masked correctness: same category	Masked correctness: exact	Factual consistency	Zlog mean |z| ↓	Zlog satisfactory	Faithfulness: supported	Case alignment: aligned	Strict truth coverage*	nDCG@8*	Retrieval cost / question	Total cost / question	Cost / run	Time / answer
Gemini 3 Flash knowledge no search: the model states the typical value	10	84.4 ± 1.5%	45.6 ± 1.4%	97.3 ± 1.5%	1.28 ± 0.04	74.6 ± 2.0%	–	18.6 ± 0.9%	–	–	$0.0007	$0.0026	$1.07	70 ± 8 s
Europe PMC + LitSense, original the retrieval as first built (no changes 1–4)	10	86.2 ± 1.5%	45.4 ± 2.5%	96.8 ± 1.2%	1.26 ± 0.05	73.0 ± 1.8%	22.7 ± 1.3%	20.0 ± 1.1%	40.4%	0.593	$0.0000	$0.0024	$0.97	123 ± 13 s
Europe PMC + LitSense, changes 1–4 medsim's current default	10	84.9 ± 1.3%	41.6 ± 2.1%	96.8 ± 1.1%	1.29 ± 0.05	73.1 ± 2.8%	35.1 ± 2.5%	20.1 ± 0.6%	64.5%	0.699	$0.0000	$0.0025	$1.03	107 ± 5 s
Exa (instant) OpenRouter web search, medical domains	10	85.0 ± 1.5%	45.9 ± 3.0%	97.2 ± 0.9%	1.25 ± 0.06	75.4 ± 2.7%	21.0 ± 1.3%	20.2 ± 0.8%	–	–	$0.0070	$0.0092	$3.80	79 ± 11 s
Parallel (basic) OpenRouter web search, medical domains	10	84.0 ± 2.2%	43.4 ± 2.6%	96.7 ± 1.0%	1.29 ± 0.06	71.9 ± 2.8%	25.4 ± 1.8%	20.3 ± 1.0%	–	–	$0.0051	$0.0074	$3.06	84 ± 6 s
Perplexity OpenRouter web search, medical domains	10	83.2 ± 2.3%	43.9 ± 2.7%	97.1 ± 0.6%	1.32 ± 0.04	72.1 ± 2.1%	59.2 ± 2.4%	18.2 ± 0.8%	–	–	$0.0051	$0.0078	$3.22	93 ± 8 s
Native Google search Gemini 3.1 Flash Lite's own search (one run)	1	82.3%	44.7%	96.7%	1.25	71.4%	23.5%	19.1%	–	–	$0.0501	$0.0511	$21.09	82 s
Native OpenAI search GPT-6 Luna's own search (one run)	1	84.4%	44.0%	95.6%	1.28	72.9%	38.3%	19.5%	–	–	$0.0253	$0.0262	$10.80	80 s
Case only (reference) no retrieval; set C questions the case answers	10	99.2 ± 0.4%	98.8 ± 0.4%	99.7 ± 0.3%	0.04 ± 0.02	98.7 ± 0.6%	–	97.7 ± 0.8%	–	–	–	$0.0002	$0.05	8 ± 1 s

* Document metrics (section 3's definitions) on stored run 2's documents, measured only for Europe PMC + LitSense. Retrieval cost: search fees and search or selection model tokens (Europe PMC and LitSense are free). Total cost includes medsim's three LLM stages. Cost per run is the mean of runs 2 and later (OpenRouter raised the DeepSeek price after run 1); the native searches ran once. Case only answers 272 set C questions and scores near 100% by construction: it is a ceiling, not a search method.

2. Earlier optimizations of the original retrieval (50-question study, 2026-09-21)

One run of 50 questions with the earlier metrics, before the repeated benchmark: the changes 1–6 to Europe PMC + LitSense and the first OpenRouter web search. Changes 1–4 became medsim's default.

Method	Relevant documents (grade ≥ 2)	Questions with a document containing the true value	Final answer close to the true value	Final answer in the true category	Retrieval cost / question	Total cost / question	Time / question
Current (original) Europe PMC + LitSense as first evaluated	45%	22%	44%	84%	$0.00000	$0.00108	87 s
Current + Europe PMC fix malformed Europe PMC replies retried	50%	38%	47%	81%	$0.00000	$0.00100	79 s
Changes 1–4 value-first ranking, animal and age filter, excerpts, new fallbacks	63%	53%	59%	81%	$0.00000	$0.00093	96 s
Changes 1–5 plus natural-language LitSense queries	62%	50%	38%	78%	$0.00000	$0.00089	102 s
Changes 1–6 plus an LLM that picks the 8 documents	59%	47%	47%	78%	$0.00091	$0.00178	196 s
OpenRouter search Exa web search tool, 8 results, medical domains	35%	41%	53%	81%	$0.00669	$0.00750	83 s
OpenRouter search, improved the same with changes 1–3 applied	48%	47%	56%	78%	$0.00685	$0.00754	83 s

Figures from results/README.md (2026-09-21). Quality is on the 44 questions where every method retrieved (32 with a hidden value).

3. The document-metric search on Europe PMC + LitSense (2026-10-16)

141 set A questions, one frozen search query per question, retrieval only, one judge (Claude Haiku 5.5). Strict truth coverage decides: a document gives a value for patients like this one and the true value lies in its range or within about 15%. Δ is paired against the baseline over the same questions, with 95% bootstrap intervals (green: excludes zero). Faithfulness (answers written on these documents and judged by the same judge) was measured for the baseline and the winner. Best: **llm_select20+epmc25+q_synonym**, strict coverage 70.2% against 60.3%.

Configuration	Strict truth coverage	Δ vs baseline	Category coverage	nDCG@8	Δ nDCG@8	Precision@8	Faithfulness	Retrieval cost / question
Baseline								
baseline medsim's default (changes 1–4)	60.3%		83.7%	0.691		41.3%	22.0%	$0.0000
Combinations								
llm_select20+epmc25+q_synonym an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion	**70.2%**	**+9.9 (+3.5 to +17.0)**	**87.9%**	**0.750**	**+0.059 (+0.042 to +0.076)**	**49.6%**	**32.6%**	**$0.0016**
llm_select20+q_synonym an LLM picks the 8 documents from the best 20; Europe PMC MeSH synonym expansion	68.8%	+8.5 (+2.8 to +14.9)	90.8%	0.743	+0.052 (+0.035 to +0.070)	48.8%	–	$0.0017
llm_select20+epmc25 an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50)	68.8%	+8.5 (+1.4 to +14.9)	87.9%	0.737	+0.046 (+0.027 to +0.065)	48.3%	–	$0.0016
Single changes								

Configuration	Strict truth coverage	Δ vs baseline	Category coverage	nDCG@8	Δ nDCG@8	Precision@8	Faithfulness	Retrieval cost / question
llm_select40 an LLM picks the 8 documents from the best 40	68.8%	+8.5 (+2.1 to +15.6)	89.4%	0.746	+0.055 (+0.030 to +0.078)	50.8%	–	$0.0030
llm_select20 an LLM picks the 8 documents from the best 20	67.4%	+7.1 (+0.7 to +14.2)	89.4%	0.749	+0.058 (+0.041 to +0.076)	49.6%	–	$0.0018
epmc25 25 Europe PMC candidates (baseline 50)	66.0%	+5.7 (-1.4 to +12.8)	85.1%	0.690	-0.001 (-0.020 to +0.017)	42.6%	–	$0.0000
q_synonym Europe PMC MeSH synonym expansion	63.8%	+3.5 (-1.4 to +8.5)	84.4%	0.690	-0.001 (-0.015 to +0.013)	42.3%	–	$0.0000
epmc_only Europe PMC only	63.1%	+2.8 (-2.8 to +8.5)	85.8%	0.665	-0.026 (-0.048 to -0.005)	40.5%	–	$0.0000
ls_sentences LitSense sentences instead of passages	63.1%	+2.8 (-4.3 to +9.9)	86.5%	0.659	-0.032 (-0.053 to -0.013)	39.3%	–	$0.0000
q_natural LitSense natural-language queries	62.4%	+2.1 (-4.3 to +8.5)	83.7%	0.684	-0.007 (-0.029 to +0.014)	41.3%	–	$0.0000
round_robin round-robin merge across sources instead of one list	62.4%	+2.1 (-4.3 to +8.5)	83.7%	0.667	-0.024 (-0.044 to -0.006)	39.6%	–	$0.0000
ls100 100 LitSense results	61.7%	+1.4 (-5.0 to +7.8)	86.5%	0.681	-0.010 (-0.028 to +0.007)	40.9%	–	$0.0000
epmc100 100 Europe PMC candidates	61.7%	+1.4 (-5.0 to +7.8)	85.8%	0.679	-0.012 (-0.028 to +0.004)	40.5%	–	$0.0000
fulltext20 20 open-access full texts fetched for excerpts (baseline 10)	61.0%	+0.7 (-5.7 to +7.1)	87.9%	0.700	+0.009 (-0.008 to +0.026)	44.7%	–	$0.0000
ls60 60 LitSense results (baseline 30)	61.0%	+0.7 (-5.7 to +7.1)	86.5%	0.681	-0.010 (-0.025 to +0.005)	41.2%	–	$0.0000
ls_no_rerank no LitSense-side rerank	61.0%	+0.7 (-5.7 to +7.1)	87.2%	0.677	-0.014 (-0.033 to +0.004)	40.5%	–	$0.0000
chars3000 3,000 characters per document (baseline 1,500)	60.3%	+0.0 (-6.4 to +6.4)	86.5%	0.677	-0.013 (-0.035 to +0.006)	41.7%	–	$0.0000
no_value_first no value-first ranking	59.6%	-0.7 (-7.1 to +5.7)	84.4%	0.683	-0.008 (-0.026 to +0.009)	41.5%	–	$0.0000
ladder_v1 the original broadening ladder (v1)	59.6%	-0.7 (-7.8 to +6.4)	85.8%	0.661	-0.029 (-0.050 to -0.010)	37.8%	–	$0.0000
q_gpt6luna openai/gpt-6-luna writes the search query	58.9%	-1.4 (-10.6 to +7.1)	87.9%	0.704	+0.014 (-0.016 to +0.040)	44.0%	–	$0.0000
no_population_filter no animal and age-group filter	56.7%	-3.5 (-9.2 to +2.1)	85.8%	0.682	-0.008 (-0.026 to +0.008)	40.7%	–	$0.0000
no_excerpts no full-text excerpts (abstracts and passages only)	39.0%	-21.3 (-29.8 to -12.8)	70.9%	0.597	-0.094 (-0.122 to -0.068)	29.7%	–	$0.0000
no_lexical_rerank no lexical rerank within each source	34.8%	-25.5 (-34.0 to -17.0)	66.0%	0.554	-0.136 (-0.164 to -0.110)	20.3%	–	$0.0000

Configuration	Strict truth coverage	Δ vs baseline	Category coverage	nDCG@8	Δ nDCG@8	Precision@8	Faithfulness	Retrieval cost / question
litsense_only LitSense only	29.8%	-30.5 (-40.4 to -21.3)	61.7%	0.576	-0.115 (-0.144 to -0.087)	23.8%	–	$0.0000

Checks: a shuffled control (each question judged on another question's documents) scores 0.0%; the baseline judged twice differs by 1.4 points; five stored benchmark runs of the baseline score 61.7 ± 3.4%. On two fresh query sets the winner gained +7.1 and +12.1 points. The single-judge faithfulness values are not comparable with section 1's panel values.